

LDL06F08A

2008 年 10 月 28 日

第 1 章

論理譜

```

{ ===== }
{ 論理譜 }
{ ===== }
logicname LDL06F08A

{ ----- }
{ ライブラリ追加 }
{ ----- }
library LDL06F06A0 { 端末 }
library LDL06E21A { BCD -> BIN 変換器 }
library LDL06F04A { BIN -> BCD 変換器 }
library LDL06E19A { 乗算器 }
library LDL06F04B { 除算器 }

{ ===== }
{ 実効譜 }
{ ===== }
entity main
{ ----- }
{ 入力 }
{ ----- }
input RESET;
input SCLK;
input YIN[4];

{ ----- }
{ 出力 }
{ ----- }
output XSEL[4];
output SEGOUT[7];
output KETAOUT[8];

output OFF;
output WAIT;

{ ----- }
{ 内部信号 }
{ ----- }
bitn term[28];
bitn change[36];
bitn data[4];
bitn add[3];
bitr take;
bitr setseq[3];
bitn changerereset;
bitn set;
bitn func[3];

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bitr calcseq[4];
bitr calcop[3];
bitn opend;
bitr regA[32];
bitr regB[32];
bitn bin[32];
bitr execop[3];
bitn changecomplete;
bitr answer[32];
bitn chgbintobcd[5];
bitn bindata[32];
bitn wp;
bitn addb[3];
bitn chgbintobcdeoc;
bitr tmpcount[3];
bitn dspbcd[4];
bitn dspbcdwp;
bitr dep[3];
bitn ep;
bitn chgbintobcdres;
bitn mul[33];
bitn mulreset;
bitn mulready;
bitn mulans[32];
bitn div[33];
bitn divreset;
bitn divready;
bitn divans[32];
bitn off;
bitr termwritestart[2];
bitr delayoff;

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{ ----- }
{  検査端子                               }
{ ----- }

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output T0P[4];   T0P=calcseq;
output T1P;      T1P=set;
output T2P[3];   T2P=func;
output T3P;      T3P=take;
output T4P[32];  T4P=bin;
output T5P[32];  T5P=regA;
output T6P[3];   T6P=calcop;
output T7P[3];   T7P=execop;
output T8P[32];  T8P=regB;
output T9P[3];   T9P=setseq;
output T10P[32]; T10P=answer;
output T11P;     T11P=chgbintobcdeoc;
output T12P[3];  T12P=addb;
output T13P[4];  T13P=dspbcd;
output T14P;     T14P=wp;
output T15P;     T15P=dspbcdwp;
output T16P;     T16P=opend;
output T17P;     T17P=mulreset;
output T18P;     T18P=mulready;
output T19P;     T19P=divreset;
output T20P;     T20P=divready;
output T21P[3];  T21P=add;
output T22P[4];  T22P=data;
output T23P;     T23P=changecomplete;
output T24P;     T24P=chgbintobcdres;
output T25P[2];  T25P=termwritestart;
output T26P;     T26P=ep;
output T27P[3];  T27P=tmpcount;
output T28P;     T28P=changereset;

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output T29P[32]; T29P=mulans;
output T30P[32]; T30P=divans;
```

```
{ ----- }
{ 出力代入 }
{ ----- }

OFF=off;

switch(calcseq)
  case 0:
    switch(termwritestart)
      case 1: WAIT=1;
      case 2: WAIT=1;
    endswitch
  default: WAIT=1;
endswitch

{ ----- }
{ 遅延 off }
{ ----- }

delayoff=off;

{ ----- }
{ 端末書き込み開始終了行程 }
{ ----- }

if (RESET)
  termwritestart=0;
else
  switch(termwritestart)
    case 0:
      switch(calcseq)
        case 0: if (off&!delayoff) termwritestart=1; endif
        default:
          if (chgbintobcdeoc) termwritestart=2; endif
      endswitch
    case 1: termwritestart=2;
    case 2: switch(tmpcount)
      case 7:
        if (ep)
          termwritestart=3;
        else
          termwritestart=termwritestart;
        endif
      default: termwritestart=termwritestart;
    endswitch
    case 3:
      switch(calcseq)
        case 4: termwritestart=0;
        case 8: termwritestart=0;
        default: termwritestart=termwritestart;
      endswitch
    endswitch
  endif

{ ----- }
{ 端末書き込み行程 }
{ ----- }

if (RESET)
  dep=0;
else
  switch(termwritestart)
    case 2:
      switch(dep)
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        case 2: dep=0;
        default: dep=dep+1;
    endswitch
endswitch
endif

{-----}
{  端末書き込み起点  }
{-----}
switch(dep)
    case 2: ep=1;
endswitch

{-----}
{  端末書き込み指標  }
{-----}
switch(termwritestart)
    case 2:
        switch(dep)
            case 0: dspbcdwp=1;
        endswitch
    endswitch

{-----}
{  除算器  }
{-----}
div=LDL06F04B(divreset,regA,regB,0);

div.33=1;

{-----}
{  除算結果  }
{-----}
divans=div.0:31;

{-----}
{  除算終了  }
{-----}
divready=div.32;

{-----}
{  除算初期化  }
{-----}
if (RESET)
    divreset=1;
else
    switch(execop)
        case 3:
            switch(calcseq)
                case 2: divreset=1;
            endswitch
        endswitch
    endswitch
endif

{-----}
{  乗算器  }
{-----}
mul=LDL06E19A(mulreset,regA,regB);

mul.33=1;

{-----}
{  乗算結果  }
{-----}

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mulans=mul.0:31;

{ ----- }
{ 乗算終了 }
{ ----- }
mulready=mul.32;

{ ----- }
{ 乗算初期化 }
{ ----- }
if (RESET)
    mulreset=1;
else
    switch(execop)
        case 2:
            switch(calcseq)
                case 2: mulreset=1;
            endswitch
        endswitch
endif

{ ----- }
{ 端末 }
{ ----- }
term=LDL06F06A0(RESET,SCLK,YIN,add,take,addb,dspbcd,dspbcdwp);

term.28=1;

off=term.27;

{ ----- }
{ キーボード駆動出力 }
{ ----- }
XSEL=term.0:3;

{ ----- }
{ 表示出力 }
{ ----- }
SEGOUT=term.4:10;

{ ----- }
{ 表示桁出力 }
{ ----- }
KETAOUT=term.11:18;

{ ----- }
{ 端末データ }
{ ----- }
data=term.19:22;

{ ----- }
{ 端末要求 }
{ ----- }
set=term.23;

{ ----- }
{ 機能キーデータ }
{ ----- }
func=term.24:26;

{ ----- }
{ BCD2 進数変換器 }
{ ----- }
change=LDL06E21A(changereset,data);

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change.36=1;

{-----}
{ 変換後バイナリ値 }
{-----}
bin=change.3:34;

{-----}
{ BCD2 進数変換桁指定 }
{-----}
add=change.0:2;

{-----}
{ BCD2 進数変換終了 }
{-----}
changecomplete=change.35;

{-----}
{ 2 進数 BCD 変換器 }
{-----}
chgbintobcd=LDL06F04A(chgbintobcdres,regA,wp,addb);

chgbintobcd.5=1;

{-----}
{ 2 進数 BCD 変換終了 }
{-----}
chgbintobcdeoc=chgbintobcd.4;

{-----}
{ 端末書き込み桁指定 }
{-----}
addb=tmpcount;

{-----}
{ 変換後 BCD 値 }
{-----}
dspbcd=chgbintobcd.0:3;

{-----}
{ 2 進数 BCD 変換器にデータを書き込む }
{-----}
switch(calcseq)
  case 5: wp=1;
endswitch

{-----}
{ 2 進数 BCD 変換初期化 }
{-----}
if (RESET)
  chgbintobcdres=1;
else
  switch(calcseq)
    case 4: chgbintobcdres=1;
    case 8: chgbintobcdres=1;
  endswitch
endif

{-----}
{ 端末要求行程 }
{-----}
if (RESET)
  setseq=0;

```



```

else
    switch(setseq)
    case 0:
        if (set) setseq=1; endif    { 端末数値確定 }
    case 1:
        if (take)
            setseq=2;                { BCD を 2 進数に変換終了 }
        else
            setseq=setseq;            { BCD を 2 進数に変換中 }
        endif
    case 2: setseq=4;
    case 4:
        if (set)
            setseq=setseq;
        else
            setseq=0;
        endif
    endswitch
endif

{ ----- }
{ 端末応答 }
{ ----- }

if (RESET)
    take=0;
else
    switch(take)
    case 0:
        if (changecomplete) take=1; endif
    case 1:
        if (changecomplete)
            take=take;
        else
            if (set)
                take=take;
            else
                take=0;
            endif
        endif
    endswitch
endif

{ ----- }
{ BCD2 進数変換器初期化 }
{ ----- }

switch(setseq)
case 1: changereset=0;
case 2: changereset=0;
default: changereset=1;
endswitch

{ ----- }
{ 主行程 }
{ ----- }

if (RESET)
    calcseq=0;
else
    switch(calcseq)
    case 0: { 数値入力を待つ }
        if (set)
            calcseq=1;
        else
            calcseq=calcseq;
        endif
    endswitch
endif

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case 1:      { BCD から 2 進数に変換中 }
  if (setseq.1)
    calcseq=2;
  else
    calcseq=calcseq;
  endif

case 2:      { 端末の応答待ち }
  if (set)
    calcseq=calcseq;
  else
    calcseq=3;
  endif

case 3:      { 計算の実行 }
  if (opend)
    calcseq=4;
  else
    calcseq=calcseq;
  endif

case 4:      { 計算の終了 }
  calcseq=5;

case 5:      { 計算結果を 2 進数から BCD に変換開始 }
  calcseq=6;
case 6:      { 計算結果を 2 進数から BCD に変換中 }
  if (chgbintobcdeoc)
    calcseq=7;
  else
    calcseq=calcseq;
  endif
case 7:      { 計算結果を 2 進数から BCD に変換終了 }
  switch(tmpcount)
    case 7:
      if (ep)
        calcseq=8;
      else
        calcseq=calcseq;
      endif
    default: calcseq=calcseq;
  endswitch
case 8:      { 計算結果 BCD 値転送終了 }
  calcseq=0;
endswitch
endif

{ ----- }
{ 端末書き込み計数 }
{ ----- }

if (RESET)
  tmpcount=0;
else
  switch(tmpcount)
    case 7:
      switch(termwritestart)
        case 3: tmpcount=0;
        default: tmpcount=tmpcount;
      endswitch
    default:
      if (ep)
        tmpcount=tmpcount+1;
      else
        switch(termwritestart)
          case 1: tmpcount=1;
          default: tmpcount=tmpcount;
        endswitch
      endif
    endswitch
  endswitch
endif

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        endswitch
    endif
endswitch
endif

{ ----- }
{ 計算の種類 }
{ ----- }

if (RESET)
    calcop=0;
else
    if (set)
        switch(func)
            case 1: calcop=1; { = }
            case 2: calcop=2; { * }
            case 3: calcop=3; { / }
            case 4: calcop=4; { + }
            case 5: calcop=5; { - }
        endswitch
    else
        calcop=calcop;
    endif
endif

{ ----- }
{ 実効計算 }
{ ----- }

if (RESET)
    execop=0;
else
    switch(calcseq)
        case 1:
            switch(execop)
                case 1: execop=calcop;
                default: execop=execop;
            endswitch
        case 4: execop=calcop;
        default: execop=execop;
    endswitch
endif

{ ----- }
{ レジスタ B }
{ ----- }

if (RESET)
    regB=0;
else
    if (setseq.1)
        regB=bin;
    else
        switch(calcseq)
            case 4: regB=0;
            default: regB=regB;
        endswitch
    endif
endif

{ ----- }
{ レジスタ A }
{ ----- }

if (RESET)
    regA=0;
else
    switch(calcseq)
        case 4:

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```

        switch(execop)
            case 0: regA=regB;
            case 2: regA=answer; { 乗算結果を記憶 }
            case 3: regA=answer; { 除算結果を記憶 }
            case 4: regA=answer; { 加算結果を記憶 }
            case 5: regA=answer; { 減算結果を記憶 }
        endswitch
        default: regA=regA;
    endswitch
endif

{ ----- }
{ 計算終了指標 }
{ ----- }

    switch(execop)
        case 0: opend=1;
        case 2: opend=mulready;
        case 3: opend=divready;
        case 4: opend=answer.32;
        case 5: opend=answer.32;
    endswitch

{ ----- }
{ 計算結果 }
{ ----- }

    switch(execop)
        case 2: answer=mulans;
        case 3: answer=divans;
        case 4: answer=regA+regB;
        case 5: answer=regA-regB;
    endswitch

ende

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{ ===== }
{ 機能実行譜 }
{ ===== }
entity sim
{ ----- }
{ 出力端子 }
{ ----- }
output RESET;
output SCLK;
output YIN[4];
output XSEL[4];
output SEGOUT[7];
output KETAOUT[8];
output OFF;
output WAIT;
output TC[12];
input simres;

{ ----- }
{ 内部信号 }
{ ----- }
bitr tc[12];
bitn node_XSEL[4];
bitn node_RESET;
bitr tp[8];
bitr takep[2];
bitn mode;
bitn node_OFF;
bitn node_WAIT;
bitn opration[3];
bitn selection[2];

{ ----- }
{ 検査端子 }
{ ----- }
output TBOP[8]; TBOP=tp;
output TB1P; TB1P=mode;
output TB2P[12]; TB2P=tc;

{ ----- }
{ 実効譜導入 }
{ ----- }
part main(node_RESET,SCLK,YIN,node_XSEL,SEGOUT,KETAOUT,node_OFF,node_WAIT)

RESET=node_RESET;
XSEL=node_XSEL;
OFF=node_OFF;
WAIT=node_WAIT;

{ ----- }
{ 検査設定 }
{ ----- }
{ 0: 表示器 1: キーボード }

mode=1;

{ /:0 -:1 *:2 +:3 }

opration=0;

{ A+B=:0 A+B+C=:1 }

```

```

    selection=0;

{ ----- }
{   検査位置   }
{ ----- }
    simres=0;
    if (!simres) tc=tc+1; endif

    TC=tc;

{ ----- }
{   初期化   }
{ ----- }
    if (tc<5) node_RESET=1; endif

{ ----- }
{   駆動 CLK   }
{ ----- }
    if (mode)
        SCLK=tc.2; { キーボード }
    else
        SCLK=tc.4; { 表示器 }
    endif

{ ----- }
{   データ読み取り起点   }
{ ----- }
    if (node_RESET)
        takep=0;
    else
        switch(takep)
            case 0: if (node_OFF) takep=1; endif
            case 1: takep=2;
            case 2:
                if (node_OFF)
                    takep=takep;
                else
                    takep=0;
                endif
            endif
        endswitch
    endif

{ ----- }
{   キー押し下げ回数   }
{ ----- }
    if (tc<10)
        tp=0;
    else
        if (takep==1)
            tp=tp+1;
        else
            tp=tp;
        endif
    endif

{ ----- }
{   キー入力   }
{ ----- }
    if (mode)
        if (selection==0)
            if (tc>10)
                if (node_OFF&!node_WAIT)
                    switch(tp)
                        case 0:

```

```

        switch(node_XSEL)
            case 1: YIN.1=1;    { 1 }
        endswitch
case 1:
    switch(node_XSEL)
        case 2: YIN.1=1;    { 2 }
    endswitch
case 2:
    switch(node_XSEL)
        case 4: YIN.1=1;    { 3 }
    endswitch
case 3:
    switch(node_XSEL)
        case 1: YIN.2=1;    { 4 }
    endswitch
case 4:
    switch(node_XSEL)
        case 2: YIN.2=1;    { 5 }
    endswitch
case 5:
    switch(node_XSEL)
        case 4: YIN.2=1;    { 6 }
    endswitch
case 6:
    switch(node_XSEL)
        case 1: YIN.3=1;    { 7 }
    endswitch
case 7:
    switch(node_XSEL)
        case 2: YIN.3=1;    { 8 }
    endswitch
case 8:
    switch(operation)
        case 0:
            switch(node_XSEL)
                case 8: YIN.2=1;    { / }
            endswitch
        case 1:
            switch(node_XSEL)
                case 4: YIN.0=1;    { - }
            endswitch
        case 2:
            switch(node_XSEL)
                case 8: YIN.3=1;    { * }
            endswitch
        case 3:
            switch(node_XSEL)
                case 8: YIN.1=1;    { + }
            endswitch
    endswitch
case 9:
    switch(node_XSEL)
        case 4: YIN.3=1;    { 9 }
    endswitch
case 10:
    switch(node_XSEL)
        case 1: YIN.0=1;    { 0 }
    endswitch
case 11:
    switch(node_XSEL)
        case 2: YIN.0=1;    { = }
        YIN.1=1;
    endswitch
case 12:
    if (tc>2000)
        switch(node_XSEL)

```

```

        case 8: YIN.0=1;    { C }
    endswitch
endif
endswitch
endif
endif
endif

if (mode)
if (selection==1)
if (tc>10)
if (node_OFF&!node_WAIT)
switch(tp)
    case 0:
        switch(node_XSEL)
            case 1: YIN.1=1;    { 1 }
        endswitch
    case 1:
        switch(node_XSEL)
            case 2: YIN.1=1;    { 2 }
        endswitch
    case 2:
        switch(node_XSEL)
            case 4: YIN.1=1;    { 3 }
        endswitch
    case 3:
        switch(node_XSEL)
            case 1: YIN.2=1;    { 4 }
        endswitch
    case 4:
        switch(opration)
            case 0:
                switch(node_XSEL)
                    case 8: YIN.2=1;    { / }
                endswitch
            case 1:
                switch(node_XSEL)
                    case 4: YIN.0=1;    { - }
                endswitch
            case 2:
                switch(node_XSEL)
                    case 8: YIN.3=1;    { * }
                endswitch
            case 3:
                switch(node_XSEL)
                    case 8: YIN.1=1;    { + }
                endswitch
        endswitch
    case 5:
        switch(node_XSEL)
            case 2: YIN.2=1;    { 5 }
        endswitch
    case 6:
        switch(node_XSEL)
            case 4: YIN.2=1;    { 6 }
        endswitch
    case 7:
        switch(node_XSEL)
            case 1: YIN.3=1;    { 7 }
        endswitch
    case 8:
        switch(node_XSEL)
            case 2: YIN.3=1;    { 8 }
        endswitch

```



```

case 9:
    switch(oprations)
    case 0:
        switch(node_XSEL)
        case 8: YIN.2=1;    { / }
        endswitch
    case 1:
        switch(node_XSEL)
        case 4: YIN.0=1;    { - }
        endswitch
    case 2:
        switch(node_XSEL)
        case 8: YIN.3=1;    { * }
        endswitch
    case 3:
        switch(node_XSEL)
        case 8: YIN.1=1;    { + }
        endswitch
    endswitch
case 10:
    switch(node_XSEL)
    case 4: YIN.3=1;    { 9 }
    endswitch
case 11:
    switch(node_XSEL)
    case 1: YIN.0=1;    { 0 }
    endswitch
case 12:
    switch(node_XSEL)
    case 1: YIN.1=1;    { 1 }
    endswitch
case 13:
    switch(node_XSEL)
    case 2: YIN.1=1;    { 2 }
    endswitch
case 14:
    switch(node_XSEL)
    case 2: YIN.0=1;    { = }
           YIN.1=1;
    endswitch
endswitch
endif
endif
endif
endif

```

```

if (!mode)
if (tc>10)
if (node_OFF&!node_WAIT)
switch(tp)
case 0:
    switch(node_XSEL)
    case 1: YIN.1=1;    { 1 }
    endswitch
case 1:
    switch(node_XSEL)
    case 2: YIN.1=1;    { 2 }
    endswitch
case 2:
    switch(node_XSEL)
    case 4: YIN.1=1;    { 3 }
    endswitch
case 3:
    switch(node_XSEL)

```

```
        case 1: YIN.2=1;    { 4 }
    endswitch
case 4:
    switch(node_XSEL)
        case 2: YIN.2=1;    { 5 }
    endswitch
case 5:
    switch(node_XSEL)
        case 4: YIN.2=1;    { 6 }
    endswitch
case 6:
    switch(node_XSEL)
        case 1: YIN.3=1;    { 7 }
    endswitch
case 7:
    switch(node_XSEL)
        case 2: YIN.3=1;    { 8 }
    endswitch
case 8:
    if (tc>2500)
        switch(node_XSEL)
            case 8: YIN.0=1;    { C }
        endswitch
    endif
endswitch
endif
endif
endif

ende

endlogic
```